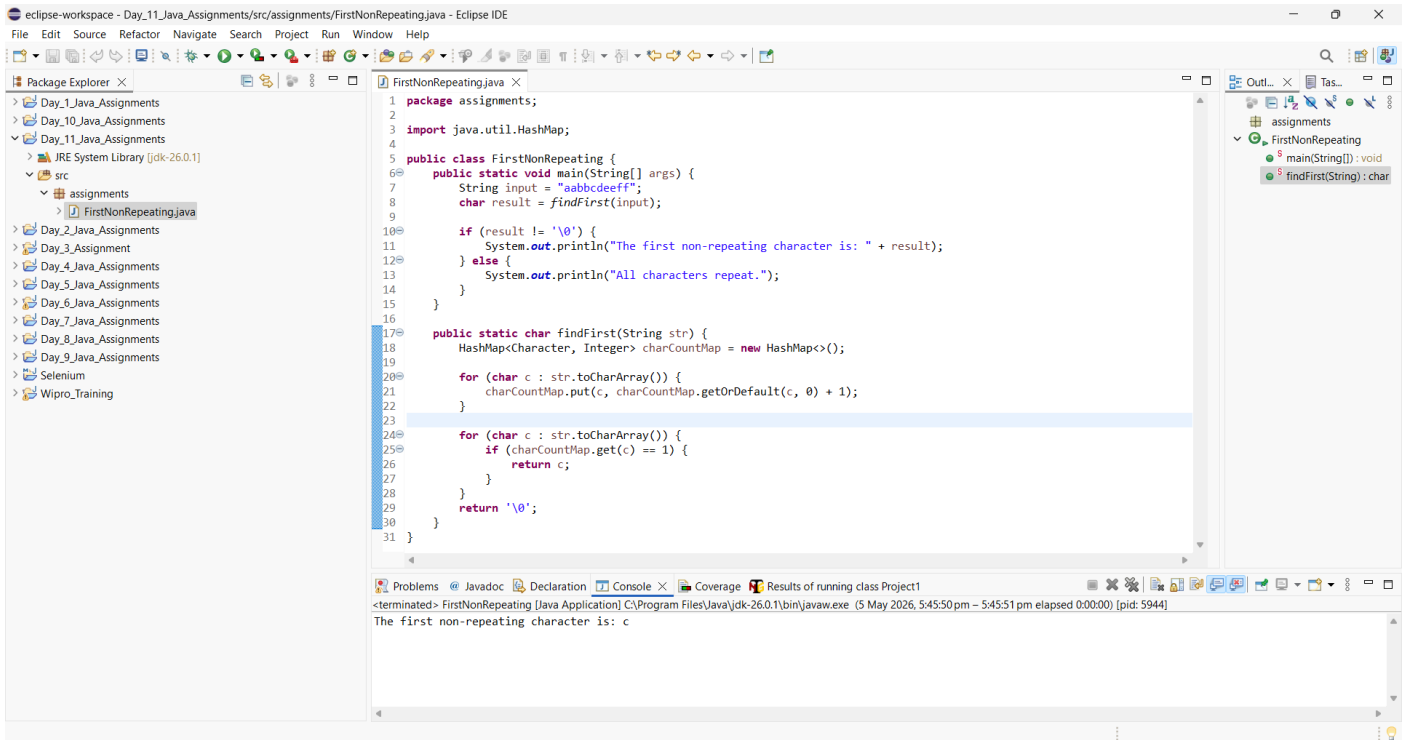


1. Find first non-repeating character in a string.



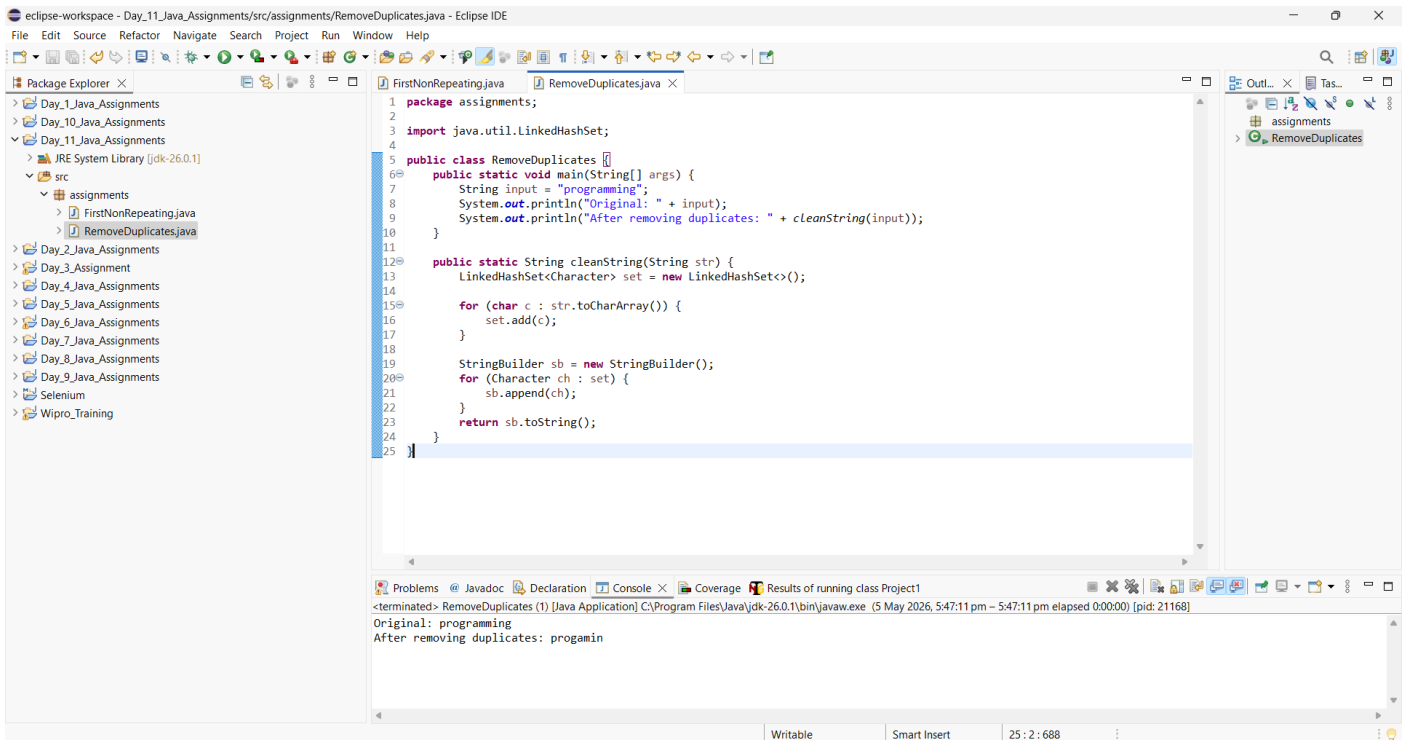
```
1 package assignments;
2
3 import java.util.HashMap;
4
5 public class FirstNonRepeating {
6     public static void main(String[] args) {
7         String input = "aabbcddeeff";
8         char result = findFirst(input);
9
10        if (result != '\0') {
11            System.out.println("The first non-repeating character is: " + result);
12        } else {
13            System.out.println("All characters repeat.");
14        }
15    }
16
17    public static char findFirst(String str) {
18        HashMap<Character, Integer> charCountMap = new HashMap<>();
19
20        for (char c : str.toCharArray()) {
21            charCountMap.put(c, charCountMap.getOrDefault(c, 0) + 1);
22        }
23
24        for (char c : str.toCharArray()) {
25            if (charCountMap.get(c) == 1) {
26                return c;
27            }
28        }
29        return '\0';
30    }
31 }
```

Problems | Javadoc | Declaration | Console | Coverage | Results of running class Project1

<terminated> FirstNonRepeating [Java Application] C:\Program Files\Java\jdk-26.0.1\bin\javaw.exe (5 May 2026, 5:45:50 pm – 5:45:51 pm elapsed 0:00:00) [pid: 5944]

The first non-repeating character is: c

2. Remove duplicate characters from string.



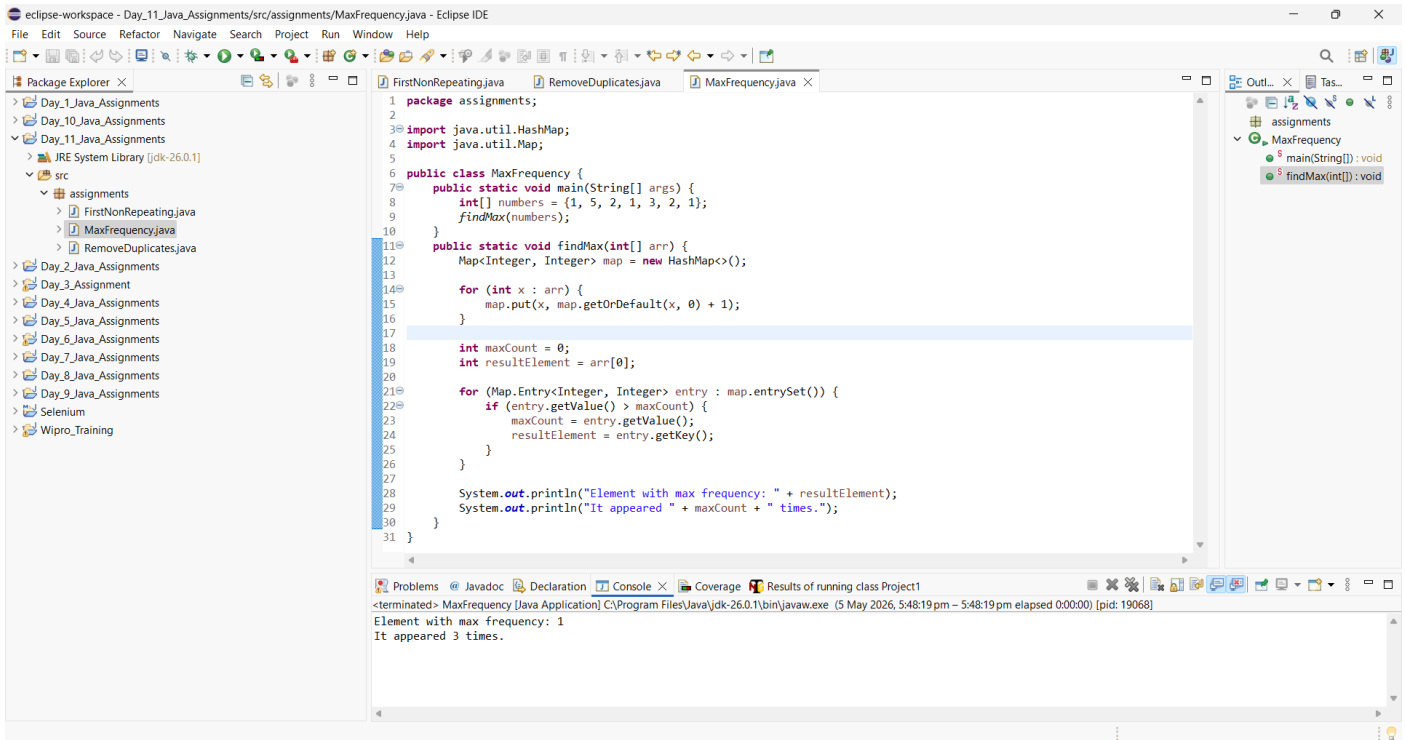
```
1 package assignments;
2
3 import java.util.LinkedHashSet;
4
5 public class RemoveDuplicates {
6     public static void main(String[] args) {
7         String input = "programming";
8         System.out.println("Original: " + input);
9         System.out.println("After removing duplicates: " + cleanString(input));
10    }
11
12    public static String cleanString(String str) {
13        LinkedHashSet<Character> set = new LinkedHashSet<>();
14
15        for (char c : str.toCharArray()) {
16            set.add(c);
17        }
18
19        StringBuilder sb = new StringBuilder();
20        for (Character ch : set) {
21            sb.append(ch);
22        }
23        return sb.toString();
24    }
25 }
```

Problems | Javadoc | Declaration | Console | Coverage | Results of running class Project1

<terminated> RemoveDuplicates (1) [Java Application] C:\Program Files\Java\jdk-26.0.1\bin\javaw.exe (5 May 2026, 5:47:11 pm – 5:47:11 pm elapsed 0:00:00) [pid: 21168]

Original: programming
After removing duplicates: progamin

3. Find element with maximum frequency.



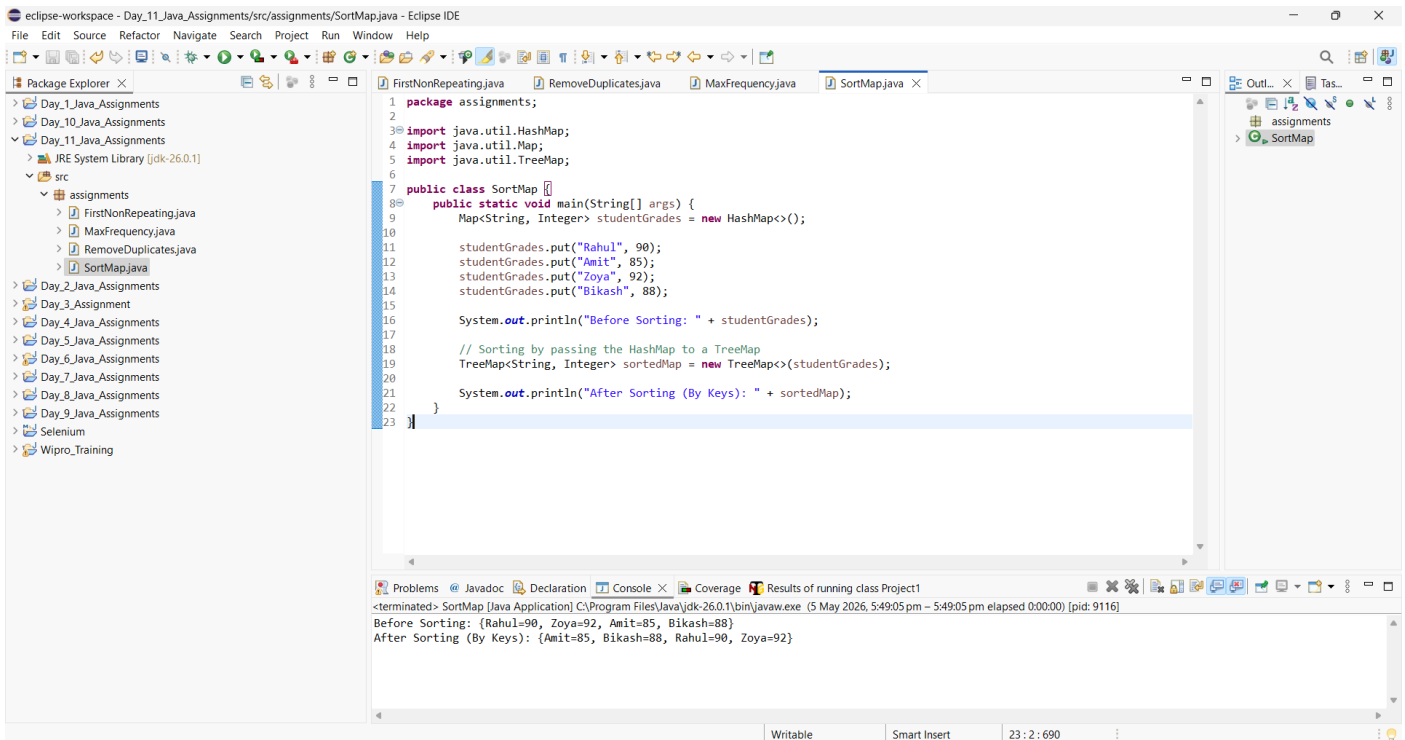
The screenshot shows the Eclipse IDE with the `MaxFrequency.java` file open. The code defines a `MaxFrequency` class with a `main` method that takes an array of integers and a `findMax` static method that uses a `HashMap` to find the element with the maximum frequency. The console output shows the result of running the class.

```
1 package assignments;
2
3 import java.util.HashMap;
4 import java.util.Map;
5
6 public class MaxFrequency {
7     public static void main(String[] args) {
8         int[] numbers = {1, 5, 2, 1, 3, 2, 1};
9         findMax(numbers);
10    }
11    public static void findMax(int[] arr) {
12        Map<Integer, Integer> map = new HashMap<>();
13
14        for (int x : arr) {
15            map.put(x, map.getOrDefault(x, 0) + 1);
16        }
17
18        int maxCount = 0;
19        int resultElement = arr[0];
20
21        for (Map.Entry<Integer, Integer> entry : map.entrySet()) {
22            if (entry.getValue() > maxCount) {
23                maxCount = entry.getValue();
24                resultElement = entry.getKey();
25            }
26        }
27
28        System.out.println("Element with max frequency: " + resultElement);
29        System.out.println("It appeared " + maxCount + " times.");
30    }
31 }
```

Console Output:

```
<terminated> MaxFrequency [Java Application] C:\Program Files\Java\jdk-26.0.1\bin\javaw.exe (5 May 2026, 5:48:19 pm - 5:48:19 pm elapsed 0:00:00) [pid: 19068]
Element with max frequency: 1
It appeared 3 times.
```

4. Sort a map by keys.



The screenshot shows the Eclipse IDE with the `SortMap.java` file open. The code defines a `SortMap` class with a `main` method that takes an array of strings and integers, creates a `HashMap`, and then sorts it using a `TreeMap`. The console output shows the result of running the class.

```
1 package assignments;
2
3 import java.util.HashMap;
4 import java.util.Map;
5 import java.util.TreeMap;
6
7 public class SortMap {
8     public static void main(String[] args) {
9         Map<String, Integer> studentGrades = new HashMap<>();
10
11         studentGrades.put("Rahul", 90);
12         studentGrades.put("Amit", 85);
13         studentGrades.put("Zoya", 92);
14         studentGrades.put("Bikash", 88);
15
16         System.out.println("Before Sorting: " + studentGrades);
17
18         // Sorting by passing the HashMap to a TreeMap
19         TreeMap<String, Integer> sortedMap = new TreeMap<>(studentGrades);
20
21         System.out.println("After Sorting (By Keys): " + sortedMap);
22     }
23 }
```

Console Output:

```
<terminated> SortMap [Java Application] C:\Program Files\Java\jdk-26.0.1\bin\javaw.exe (5 May 2026, 5:49:05 pm - 5:49:05 pm elapsed 0:00:00) [pid: 9116]
Before Sorting: {Rahul=90, Zoya=92, Amit=85, Bikash=88}
After Sorting (By Keys): {Amit=85, Bikash=88, Rahul=90, Zoya=92}
```